

Ahmed Ayman Al-Hofy

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Summary

Data Science & AI Engineer with a background in Information Systems, specializing in RAG, NLP, and data automation. Experienced in building **end-to-end AI agents** and **predictive models** that solve real-world problems. Passionate about bridging the gap between **raw data** and **impactful AI-driven products**.

Education

Bachelor's Degree, Faculty of Computers & Artificial Intelligence, University of Sadat City Sep 2022 – Jul 2026

CGPA: 3.43/4.0 (Excellent with Honors) | Ranked **8th** in the IS Department and **13th** across the Faculty, with an **A+** graduation project.

Experience & Training

Associate Salesforce Developer | Arcsen · Internship Qesm El Maadi, Cairo, Egypt | Jul 2026 – Aug 2026

- Selected for the Arcsen Academy training program and won the Best Team award.
- Built hands-on expertise in Salesforce through data modeling, security, Flows, Apex, and Lightning Web Components (LWC).
- Graduation Project: Engineered an end-to-end Salesforce travel operations platform integrating REST APIs, real-time LWCs, automated payments, custom data models, and role-based security to streamline bookings and prevent conflicts, while building a profitability reporting dashboard to track revenue and margins and improve decision visibility.

AI Engineer | Konecta · Internship New Cairo, Egypt | Jul 2025 – Jan 2026

- Collected, cleaned, and structured data using Python and web scraping tools to support reliable analysis and modeling.
- Built and optimized machine learning models (classification, clustering, image recognition) and implemented RAG-based chatbots with embeddings, document indexing, and multi-turn conversations.
- Automated data and AI workflows using n8n, improving efficiency and reducing manual processing.

Data scientist | Digital Egypt Pioneers Initiative - DEPI · Internship Alexandria, Egypt | Oct 2024 – May 2025

- Learned to analyze data using Python, SQL, and analytics tools to uncover trends, opportunities, and actionable business insights.
- Learned to build, deploy, and track ML models using Python, Scikit-learn, and MLflow, enhanced with AI and prompt engineering for seamless integration.

Intern | CIB Egypt · Internship Cairo, Egypt | Jul 2024 - Jul 2024

- Developed data, financial, and entrepreneurial skills to strengthen problem-solving and decision-making.

Skills

Python (ML/DS) | SQL | HTML/CSS | LangChain | LangGraph | LLMs (Hugging Face) | Prompt Engineering | RAG | Matplotlib | Seaborn | Scikit-learn | TensorFlow | CNN | Web Scraping (Playwright, BeautifulSoup) | AWS | Google Cloud Platform (GCP) | Vector Databases | Stats | Tableau | Power BI | Excel | Docker | Git | OOP | Problem-Solving | Communication

Projects

AI Real Estate Chat Consultant GitHub | Live Demo

- Developed an AI property consultant using a RAG pipeline (LangChain, LangSmith) to enable efficient querying of nearly 9,000 properties and 600+ blogs across 1,700+ compounds and 40+ locations, optimizing system latency and operational costs.
- Engineered an intelligent routing system utilizing Vector DB semantic search and LLM-based filtering to deliver precise, high-intent property recommendations and automated market comparisons.
- Built a scalable full-stack platform (FastAPI, Docker, Next.js) featuring an XGBoost model trained on scraped listings to provide real-time property price predictions and market insights.

Book Semantic Search & Recommendation System GitHub | Live Demo

- Improved recommendation relevance over keyword search by building a semantic search engine using Sentence-Transformers embeddings and Zero-Shot classification.
- Enabled scalable multi-environment deployment by developing a full-stack Next.js and FastAPI application with Docker containerization and GitHub Actions CI/CD workflows.
- Enhanced book discovery across 7,000+ titles by applying EDA and NLP-based sentiment analysis to generate custom mood and semantic tags.

Colon Cancer Histopathology Classification GitHub | Live Demo

- Built a deep learning CNN-based colon histopathology classifier that achieved 99.19% test accuracy and 0.99 F1-score by leveraging EfficientNetB3 transfer learning and optimized TensorFlow pipelines with only 197K trainable parameters (1.8% of total).

Customer Segmentation

[GitHub](#) | [Live Demo](#)

- Segmented wholesale customers using K-Means, PCA, and rigorous evaluation to define six actionable profiles, driving targeted marketing and inventory optimization.

Automated Document Processing

[GitHub](#) | [Live Demo](#)

- Built an AI document processor using OCR, LangChain, and LLM to extract structured data, cutting manual work by 80%; deployed a Streamlit app with MongoDB search for real-time querying.

Courses And Certifications

• AI Engineer for Data Scientists Associate Issued by Data Camp	Apr 2026
• SAP Certified Associate - SAP Generative AI Developer Issued by SAP	Sep 2025
• AWS Certified AI Practitioner Issued by Amazon Web Services Training and Certification	Apr 2025
• Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate Issued by Oracle	Jun 2025
• Data Scientist Associate Issued by Data Camp	Aug 2024
• IBM SkillsBuild Data Analytics Certificate Issued by IBM SkillsBuild	Jun 2025
• AWS Academy Graduate - AWS Academy Machine Learning for Natural Language Processing	Aug 2024
• AWS Academy Graduate - AWS Academy Cloud Foundations	Jun 2024
• AWS Academy Graduate - AWS Academy Data Engineering	Jun 2024
• Applied Data Science with Python - Level 2 Issued By IBM	May 2024
• Google Data Analytics Professional Certificate Issued by Coursera	Jul 2024
• The Machine Learning Process A-Z – Issued By 365 Data Science	Nov 2024
• McKinsey Forward Program - Issued by McKinsey & Company	Dec 2024